

Pixel to Camera angle

a. Pusat (Center of Image)

$$\begin{array}{l|l} x_c \text{ (x pixel)} & x_c = \frac{w}{2} \\ y_c \text{ (y pixel)} & y_c = \frac{h}{2} \end{array}$$

b. Offset from center.

$$\Delta x = x - x_c$$

$$\Delta y = y - y_c$$

c. Subst per pixel

$$FOV_{\text{img}} = \frac{FOV_{\text{sensor}}}{180}$$

$$\Delta x = \frac{FOV}{w}$$

$$\Delta y = \frac{FOV}{h}$$

d. Subst each target from camera.

$$\theta_x = \Delta x \cdot \Delta x$$

$$\theta_y = \Delta y \cdot \Delta y$$

Angle to Distance from Ground

$$X = h \cdot \tan(\theta_x)$$

X: each horizontal (front-back)

$$Y = h \cdot \tan(\theta_y)$$

Y: each vertical (up-down)

Distance to lat & lon

$$R = 6.371.000 \text{ m (Earth's radius)}$$

$$\Delta \text{lat} = \frac{X}{R}$$

$$\Delta \text{lon} = \frac{Y}{R \cdot \cos(\text{lat})}$$

coordinates.

$$\text{lat} = \text{lat}_0 + \Delta \text{lat}$$

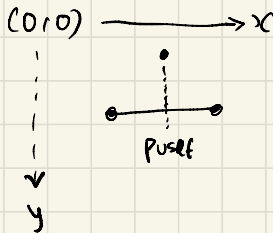
$$\text{lon} = \text{lon}_0 + \Delta \text{lon}$$

(radian)

Convert:

$$(\text{radian}) \times \frac{180}{\pi}$$

Illustrasi



Iran AH.

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